PROJECT REPORT

Socratica – Emotion-Aware Learning Companion

AI-Driven Emotion Detection & Personalized Learning Support Platform

Submitted by

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Project Team:

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Palak Agarwal – Environment & Setup
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Repository: github.com/justphase/Emotion-detector-skill-wallet
Live Demo: huggingface.co/spaces/crazyaadi/socratica-companion
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1. Introduction

Socratica is an AI-powered learning companion designed to understand a student's emotional state while they study and to respond with personalized, empathetic guidance. The platform combines emotion classification from free-text input with generative AI to deliver real-time academic support that adapts to how a learner is actually feeling – rather than assuming every learner is equally engaged, confident, or focused.

The system detects emotions such as boredom, confidence, confusion, curiosity, and frustration using two complementary deep learning approaches – a fine-tuned RoBERTa transformer and a custom-trained BiLSTM network – and uses the detected emotional signal to trigger contextual, field-aware guidance generated through the Google Gemini API, with a graceful template-based fallback when AI generation is unavailable.

1.1 Motivation

Traditional e-learning platforms treat every learner interaction identically, regardless of the learner's emotional or cognitive state. A student who is frustrated needs a very different kind of support than one who is merely bored or one who is confidently progressing. By detecting emotion directly from a learner's own words, Socratica aims to close this gap and make digital learning support more responsive and human-centered.

1.2 Objectives

- Detect a learner's primary and secondary emotional states from free-text input in real time.
- Compare the performance and practicality of a pre-trained transformer model (RoBERTa) against a lightweight custom-trained BiLSTM model.
- Generate contextual, empathetic, field-aware learning guidance using the Gemini API.
- Log sessions and present emotion analytics to help learners track their patterns over time.
- Package the entire system behind a FastAPI backend with a responsive web UI and clear REST endpoints.

2. Technology Stack

The project is built using a Python-based backend with a lightweight JavaScript/HTML/CSS frontend. The stack was chosen to balance rapid prototyping with production-realistic deployment.

Category	Technology
Backend	FastAPI, Uvicorn
Frontend	HTML, CSS, JavaScript
AI / Generation	Google Gemini API
Deep Learning	PyTorch
NLP Models	RoBERTa, BiLSTM
Charts / Analytics	Chart.js
Containerization	Docker

3. Project Structure

The repository is organized as follows:

```
Emotion-detector-skill-wallet/   main.py              - FastAPI application entry point & routes   bert_model.py        - RoBERTa-based emotion classifier   bilstm_model.py      - Custom BiLSTM emotion classifier   train_bilstm.py      - Training script for the BiLSTM model   requirements.txt     - Python dependencies   Dockerfile           - Container build definition   static/              - Frontend assets (CSS, JS)   templates/           - HTML templates   utils/               - Shared helper utilities
```

4. System Architecture & Workflow

The application follows a linear pipeline that begins with raw learner text and ends with a logged, analyzable session. The learner submits free-text input describing how they feel or what they are working on. This text is passed to both the RoBERTa and BiLSTM emotion classifiers, which independently predict the learner's primary emotion and, where confidence for a secondary emotion is 15% or higher, a secondary emotion as well. The detected emotional state is then used to prompt the Gemini API to generate tailored, field-aware learning guidance; if the Gemini API is unavailable, the system falls back to a template-based response so the learner is never left without guidance. Finally, the session – including detected emotion, model outputs, and generated guidance – is logged and surfaced on an analytics dashboard for the learner to review their patterns over time.

This produces the following high-level workflow:

```
User Input     -> Emotion Detection (RoBERTa + BiLSTM)     -> Gemini AI Response Generation     -> Session Logging     -> Analytics Dashboard
```

5. Key Features

- Emotion detection using two independent AI models (RoBERTa and BiLSTM).
- Side-by-side comparison of RoBERTa and BiLSTM predictions for the same input.
- AI-generated, field-aware learning guidance powered by Gemini.
- Emotion analytics dashboard for tracking patterns over time.
- Persistent session history logging.
- FastAPI backend exposing clean REST APIs, documented via Swagger.
- Responsive, modern web UI.

6. Emotion Categories

The model classifies learner input into the following primary emotion categories, which map to learning-relevant cognitive/affective states:

Emotion	Interpretation in a Learning Context
Bored	Low engagement; content may be too easy or repetitive.
Confident	Learner is progressing well and understands the material.

Emotion	Interpretation in a Learning Context
Confused	Learner is struggling to follow a concept.
Curious	Learner is engaged and motivated to explore further.
Frustrated	Learner is facing difficulty and may need direct support.

Secondary emotions are also reported whenever the model's confidence for them reaches 15% or higher, allowing the system to reflect mixed emotional states rather than forcing a single label.

7. API Endpoints

Method	Endpoint	Purpose
GET	/	Home page
POST	/predict	Predict emotion from input text
POST	/generate	Generate AI-based learning response
POST	/log	Save a session record
GET	/sessions	Retrieve analytics data

8. Installation & Setup

The project can be set up locally with the following steps:

```
git clone https://github.com/justphase/Emotion-detector-skill-wallet.git cd Emotion-detector-skill-wallet python -m venv venv venv\Scripts\activate pip install -r requirements.txt  # create a .env file containing: GEMINI_API_KEY=YOUR_API_KEY  python main.py
```

Once running, the application is available at http://localhost:8000, with interactive Swagger API documentation at http://localhost:8000/docs. A Dockerfile is also provided for containerized deployment.

9. Team & Individual Contributions

Team Member	Role / Contribution
Aaditya Mishra	AI Models – design, training, and evaluation of the RoBERTa and BiLSTM emotion classifiers
Dhruv Sain	Backend APIs – FastAPI routes and service integration
Zenul Aabedeen Khan	Frontend – responsive UI and dashboard
Palak Agarwal	Environment & Setup – dependency management, Docker configuration
Priya Sharma	Documentation – project write-up and usage guides

10. Future Improvements

- User authentication for personalized, secure sessions.
- Persistent database support in place of file/session-based logging.
- Cloud deployment for scalable, production-grade hosting.
- Further optimization of the BiLSTM model for accuracy and latency.
- Expansion to a broader and more fine-grained set of emotion classes.

11. Conclusion

The AI-Driven Emotion Detection & Personalized Learning Support Platform successfully demonstrates how deep learning-based emotion classification can be combined with generative AI to deliver real-time, empathetic academic support. By comparing a pre-trained transformer model (RoBERTa) against a lightweight custom-trained BiLSTM, the platform highlights the trade-off between accuracy and training cost – a key consideration in real-world machine learning deployment.

The system detects both primary and secondary (mixed) emotional states from a learner's free-text input, and uses this signal to generate contextual, field-aware guidance via the Gemini API, with a graceful template-based fallback when AI generation is unavailable. Session logging and analytics further support the platform's goal of enabling continuous learning and long-term insight into a learner's emotional patterns over time.

This project reinforced practical skills across the machine learning lifecycle – data preparation, model comparison, prompt engineering, and full-stack integration – while working under real time constraints, mirroring the kind of trade-offs made in production AI systems.

12. References & Links

- Live Demo: https://huggingface.co/spaces/crazyaadi/socratica-companion
- GitHub Repository: https://github.com/justphase/Emotion-detector-skill-wallet
- Dataset Used: GoEmotions (Hugging Face) – https://huggingface.co/datasets/go_emotions
- Pre-trained Model: RoBERTa fine-tuned on GoEmotions – https://huggingface.co/SamLowe/roberta-base-go_emotions
- Gemini API Documentation: https://ai.google.dev/gemini-api/docs